

A

MAJOR PROJECT REPORT

ON

**SILENT EMOTIONS : AI- DRIVEN LIP INTERPRETATION
AND EMOTION PERCEPTION**

*Submitted in partial fulfillment of the requirement
for the award of the degree of*

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence & Machine Learning)

BY

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Under the esteemed guidance of

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(A UGC Autonomous Institution, Approved by AICTE, Accredited by NBA & NAAC-A Grade, Affiliated to JNTUH)

VIGNANA BHARATHI
Institute of Technology

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CERTIFICATE

*This is to certify that the major project titled **SILENT EMOTIONS – AI Driven Lip Interpretation and Emotion Perception** submitted by **P. Nikitha Reddy (21P61A66D6)** in B.Tech. IV-II semester Computer Science & Engineering (Artificial Intelligence & Machine Learning) is a record of the Bonafide work carried out by her.*

The results embodied in this report have not been submitted to any other University for the award of any degree.

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ABSTRACT

The growing need for assistive technologies for individuals with speech impairments due to conditions like **aphonia, dysphonia, Parkinson's disease, and amyotrophic lateral sclerosis (ALS)** underscores the importance of developing accessible, non-verbal communication systems. Conventional solutions either rely heavily on audio input or require complex hardware, limiting their feasibility in real-time and resource-constrained environments. In response to these limitations, this project proposes an AI-powered system titled "**Silent Emotions: AI-Driven Lip Interpretation and Emotion Perception**" that provides a silent, visual-based communication method through lip reading and facial emotion analysis. The system comprises two independently trained deep learning modules. The first module performs **lip reading** by analyzing silent video frames of lip movements using a **ResNet-50** feature extractor followed by a **BiLSTM-GRU hybrid** for temporal sequence modeling. The second module performs **emotion recognition** by processing facial expressions with a similar ResNet-based architecture and sequence model. A **Late Fusion strategy** integrates the outputs of both models using **Weighted Sum Fusion**, enhancing prediction accuracy by balancing modality confidence scores.

Preprocessing involves facial landmark detection, lip region extraction, and efficient frame handling using integral images. The lip reading module is trained on **LRS3**, while the emotion recognition module uses **RAVDESS**. Real-time testing is enabled via deployment using **Flask and FastAPI**, offering video input, prediction display, and modular outputs. Initial evaluations confirm strong real-time performance, and future development will focus on deploying the system to edge devices. This project demonstrates the power of deep learning in enabling accessible, non-verbal communication for individuals with speech-related disorders.

Keywords: Artificial Intelligence, Deep Learning, Human Computer Interaction (HCI), Assistive technology, Weighted Sum Fusion, BiLSTM, GRU, ResNet-50, Multimodal Fusion, Silent Communication.

DEPARTMENT OF
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VISION

To achieve global standards of quality in technical education with the help of advanced resources and automated tools to bridge the gap between industry and academia.

MISSION

- Build the students technically competent on global arena through effective teaching learning process and world-class infrastructure.
- Inculcate professional ethics, societal concerns, technical skills and life-long learning to succeed in multidisciplinary fields.
- Establish competency center in the field of Artificial Intelligence and Machine Learning with the collaboration of industry and innovative research.

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PEO 1: Domain Knowledge: Impart strong foundation in basic sciences, Mathematics, Engineering and emerging areas by Advanced tools and Technologies.

PEO 2: Professional Employment: Develop Professional skills that prepare them for immediate employment in industry, government, entrepreneurship and R&D.

PEO 3: Higher Degrees: Motivation to pursue higher studies and acquire masters and research.

PEO4: Engineering Citizenship: Communicate and work effectively, engage in team work, achieve professional advancement, exhibit leadership skills, and ethical attitude with a sense of social responsibility.

PEO 5: Lifelong Learning: Lead edge of the industrial engineering discipline and respond to challenges of an ever-changing environment with the most current knowledge and technology.

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- 2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
- 9. Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective Presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary Environments.

12. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Develop Artificial Intelligence (AI) based software Applications/solutions as per the needs of industry and society.

PSO2: Adopt and learn emerging technologies in Artificial Intelligence, Machine Learning and Deep Learning by establishing Center of excellence.

PSO3: Apply evolutionary changes and intelligent methods of Artificial Intelligence, Machine Learning in software project development by using open-ended programming environments.

Course Objective

1. Identify and compare technical and practical issues related to the area of course specialization.
2. Design and implement projects, including several systems to solve engineering challenges and meet specified requirements.
3. Prepare a well-organized report employing elements of technical writing and critical thinking.
4. Demonstrate the ability to describe, interpret and analyze technical issues and develop competence in presenting.
5. Outline a notated bibliography of research demonstrating scholarly skills.

Course Outcomes

1. Describe fundamental concepts and principles related to projects.
2. Demonstrate how systems operate, including the relationship between hardware and software components.
3. Apply knowledge of programming techniques to design and implement solutions for specific problems.
4. Develop and analyze models for providing solutions to technical problems.
5. Adequate documentation, presentation and visual communication with ethical considerations.

CO–PO Mapping:

PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO															
CO1	2	3	1	2	-	1	1	2	3	-	2	3	1	-	1
CO2	2	3	1	2	3	1	1	2	3	3	2	3	3	-	2
CO3	3	3	3	3	3	2	2	3	3	-	2	3	3	2	3
CO4	2	3	3	3	3	2	1	2	3	-	3	3	-	3	3
CO5	-	-	-	-	1	3	2	3	3	3	-	1	-	-	3

Project Objectives

To develop a deep learning-based system for silent speech interpretation through lip reading using a ResNet-50 feature extractor and BiLSTM-GRU sequence model.

To design and implement an emotion recognition module using facial expression analysis and temporal modeling for non-verbal emotional context detection.

To integrate both modules using a Late Fusion strategy with Weighted Sum Fusion for enhanced communication accuracy.

Project Outcomes

Students will be able to apply deep learning techniques, such as ResNet, BiLSTM, and GRU, to solve real-world problems in assistive communication and emotion perception.

Gain practical experience in integrating AI tools with web-based platforms for user-friendly deployment.

Understand the technical, accessibility, and ethical challenges involved in developing AI solutions for individuals with speech and neurological impairments.

Project Mapping

PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
PRO															
PRO 1	3	3	2	2	3	2	1	1	3	2	1	3	3	2	2
PRO 2	3	3	3	3	2	1	2	2	3	3	2	3	3	3	3
PRO 3	3	3	3	3	3	2	2	2	3	3	2	3	3	3	3
PRO 4	2	3	2	2	2	1	2	2	3	3	2	3	2	2	2
PRO 5	2	2	2	1	2	1	1	2	2	3	1	3	2	2	3

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1. INTRODUCTION

Communication is a fundamental human need, yet millions of individuals worldwide struggle with speech-related impairments due to conditions such as **aphonia**, **dysphonia**, **Parkinson's disease**, and **amyotrophic lateral sclerosis (ALS)**. These conditions often result in the partial or complete loss of vocal function, significantly impacting a person's ability to interact socially, pursue education, or perform daily tasks. Traditional communication aids, such as text-to-speech systems or manual input devices, are often limited by speed, accessibility, and the lack of natural expressiveness.

With advancements in **Artificial Intelligence (AI)** and **Deep Learning (DL)**, there is now potential to build non-verbal communication systems that are faster, more intuitive, and inclusive. In particular, AI models are capable of extracting intricate temporal and spatial features from video data, making them ideal for interpreting **lip movements** and **facial expressions**—two key components of non-verbal communication. Inspired by the **McGurk effect**, which demonstrates how visual input influences speech perception, this project aims to leverage deep learning to simulate similar multisensory processing for individuals unable to speak.

This project proposes a **dual-module AI system** titled *"Silent Emotions: AI-Driven Lip Interpretation and Emotion Perception"*. The system includes a **lip-reading module** that analyzes silent video input using a **ResNet-50** feature extractor and a **BiLSTM-GRU sequence model**, and an **emotion recognition module** that identifies emotional states through facial expression analysis. These two modules are integrated using a **Late Fusion strategy with Weighted Sum Fusion**, enhancing the system's ability to generate accurate, expressive communication outputs.

The system is designed with a modular architecture that enables real-time testing and seamless demonstration of lip reading and emotion recognition capabilities. Its structure supports future adaptability for integration into web-based or edge-based platforms. By delivering a fast, non-invasive, and expressive communication interface, this project contributes meaningfully to the development of AI-driven assistive tools—paving the way for more inclusive and accessible interaction technologies for individuals with speech impairments.

1.1 Objective :

The main objective of this project, titled "**Silent Emotions: AI-Driven Lip Interpretation and Emotion Perception**," is to design and develop a deep learning-based, non-verbal communication system that enhances Human-Computer Interaction (HCI) for individuals affected by speech impairments such as **aphonia**, **dysphonia**, **Parkinson's disease**, and **amyotrophic lateral sclerosis (ALS)**. The project aims to address the limitations of conventional speech-based systems by offering a fast, accessible, and expressive alternative that relies solely on visual cues.

The first key objective is to build an accurate and efficient **lip reading model** using a combination of **ResNet-50** for spatial feature extraction and a **BiLSTM-GRU hybrid network** for sequence learning. This model is designed to interpret lip movements from silent video input and provide meaningful textual predictions, enabling non-verbal speech communication.

The second objective is to implement an **emotion recognition model** that analyzes facial expressions over time to classify emotional states such as happiness, sadness, anger, and neutrality. The model uses similar deep learning techniques for visual feature extraction and temporal modeling to provide emotional context alongside lip reading results.

Another critical objective is to integrate both models using a Late Fusion strategy, specifically Weighted Sum Fusion, to enhance the system's overall prediction accuracy and expressiveness. This multimodal approach ensures that both linguistic and emotional cues contribute meaningfully to the final output.

The system architecture supports **real-time performance** and modular testing, making it suitable for future deployment in lightweight applications. The project ultimately aims to create a scalable, non-invasive, and inclusive AI solution that empowers individuals with speech impairments, promotes digital accessibility, and demonstrates the transformative role of artificial intelligence in assistive technologies.

1.2 Scope :

The project titled "**Silent Emotions: AI-Driven Lip Interpretation and Emotion Perception**" addresses the vital need for accessible and expressive communication solutions for individuals with speech impairments. Rooted in the belief that artificial intelligence can bridge human limitations, the scope of this project includes the development of a real-time, non-verbal interaction system that relies entirely on visual data. It seeks to empower those affected by conditions like **aphonia**, **dysphonia**, **Parkinson's disease**, and **ALS** by providing an intelligent and intuitive communication interface.

The scope of the project encompasses the design, training, and integration of two advanced deep learning modules:

1.2.1 Lip Interpretation Module:

This module interprets lip movements from silent video sequences to reconstruct spoken content. It uses a **ResNet-50** model for spatial feature extraction and a **BiLSTM-GRU hybrid network** for capturing temporal dependencies. The model is trained on the **LRS3** dataset and aims to translate visual speech into accurate text output.

1.2.2 Emotion Recognition Module:

This component focuses on identifying emotional states such as happiness, sadness, anger, or neutrality based on facial expressions. Using the **RAVDESS** dataset, it extracts facial features with **ResNet-50** and models temporal variations with sequence networks, enhancing the expressive output of the system.

Additionally, the project supports real-time demonstration with modular architecture, allowing future integration into lightweight applications. The system is designed to function efficiently without relying on high-end hardware, emphasizing accessibility and scalability.

The broader scope includes the use of Late Fusion with Weighted Sum Fusion to combine outputs from both modules, resulting in enriched, context-aware communication. The passion behind this project lies in its potential to redefine assistive technologies, promote digital inclusivity, and extend AI's benefits to underserved populations.

Future directions involve deploying the system on edge devices, expanding it to support sign language and gesture recognition, and integrating it into mobile platforms to enable on-the-go accessibility.

In essence, this project delivers a scalable and impactful solution that showcases the transformative role of AI in advancing assistive communication technologies.

2. PROBLEM STATEMENT

Millions of individuals worldwide suffer from speech impairments caused by conditions such as **aphonia**, **dysphonia**, **Parkinson’s disease**, and **amyotrophic lateral sclerosis (ALS)**. These disorders severely limit a person’s ability to communicate verbally, which affects their independence, social interaction, mental well-being, and overall quality of life. Despite advancements in assistive technologies, existing communication aids often lack natural expressiveness, require significant manual input, or rely heavily on audio, rendering them ineffective for those with complete vocal loss.

- Traditional communication systems for the speech-impaired—such as text-based applications, physical input devices, or speech synthesizers—tend to be slow, non-intuitive, and unsuitable for real-time interaction.
- Moreover, most solutions fail to convey **emotional context**, which is a vital part of human communication, especially in caregiving, education, and social environments.

These limitations create a significant communication barrier for affected individuals, particularly in low-resource settings where specialized equipment and advanced therapies are inaccessible.

There is a **pressing need for a fast, expressive, and non-verbal communication system** that requires minimal physical input and captures both spoken content and emotional intent through **visual cues** alone. The solution must be scalable, easy to use, and capable of functioning in real-time without requiring high-end hardware.

This project aims to address these challenges by developing an **AI-powered lip reading and emotion recognition system** that enables seamless, silent, and expressive communication—contributing to greater digital inclusivity and improved quality of life for individuals with speech impairments.

3. LITERATURE REVIEW

Effective communication for individuals with speech impairments remains a major challenge in assistive technology. Traditional approaches—such as text-based input devices, eye-tracking systems, or manual speech aids—often lack natural expressiveness, are time-consuming, and fail to support real-time interaction. Moreover, these systems rarely convey emotional context, which is essential in human communication. With the rise of Artificial Intelligence (AI) and deep learning, new avenues have emerged that offer scalable, real-time, and expressive alternatives for non-verbal communication.

In recent years, **lip reading systems** have gained traction due to advances in deep learning. Studies have employed **Convolutional Neural Networks (CNNs)** for spatial feature extraction and **Recurrent Neural Networks (RNNs)** like **BiLSTM** and **GRU** for modeling the temporal dynamics of lip movements. Work by Assael et al. and Chung et al. on datasets such as **LRS2** and **LRS3** has shown promising accuracy in predicting spoken words from silent video sequences. Additionally, the **CTC loss function** has been widely adopted to map input sequences to unaligned output transcripts effectively.

In parallel, **facial emotion recognition** has advanced with models trained on datasets such as **RAVDESS**, **FER2013**, and **CK+**. Here, CNNs like **ResNet-50** and **Vision Transformers (ViTs)** have been used to extract facial features, while temporal models help capture dynamic expressions. Studies have explored the fusion of visual cues for multi-modal sentiment analysis, demonstrating improved robustness and contextual understanding.

These advancements indicate that a combination of lip reading and facial emotion recognition can provide a more expressive and effective assistive system. However, fusion strategies, particularly **Late Fusion with Weighted Sum**, remain underexplored in real-time, lightweight systems, presenting an opportunity for innovation in assistive communication tools.

Despite these advancements, existing systems present several limitations:

- ❖ Most focus on either lip reading or emotion recognition, but not both in an integrated manner.
- ❖ Many rely on high-end GPUs and complex infrastructure, limiting real-time use in resource-constrained settings.
- ❖ User interfaces are often designed for research purposes rather than real-world usability.
- ❖ Fusion techniques combining linguistic and emotional cues are underutilized or remain overly complex

Conclusion

This project bridges a critical gap in assistive communication by integrating advanced AI models into a single, non-verbal system. By addressing cost, usability, and accessibility, it presents a promising solution for real-time interaction among speech-impaired individuals, especially in resource-constrained settings.

4. PROPOSED SOLUTION

The proposed system is an AI-powered, dual-module solution designed to enable silent and expressive communication through visual cues alone. It focuses on individuals with speech impairments due to conditions like **aphonia**, **dysphonia**, **Parkinson's disease**, and **ALS**, offering a non-invasive, real-time alternative to traditional speech aids.

The first module performs **lip reading** by processing video frames using **ResNet-50** for spatial feature extraction, followed by a **BiLSTM-GRU hybrid network** for sequence modeling. This combination allows the system to accurately interpret visual speech patterns over time.

The second module handles **emotion recognition** by analyzing facial expressions using the same feature extraction and temporal learning strategy, identifying emotional states such as happiness, sadness, anger, and neutrality.

Outputs from both modules are integrated using a **Late Fusion strategy** with **Weighted Sum Fusion**, which ensures that both linguistic and emotional data contribute meaningfully to the final interpretation.

Designed with flexibility in mind, the system supports real-time testing and is optimized for low-resource environments. Its modular architecture allows for future deployment in web-based or mobile applications, making it a scalable, accessible, and impactful assistive communication tool.

4.1 Methodology :

The methodology involves designing a dual-module deep learning system for visual speech interpretation and emotion recognition. The **lip reading module** uses **ResNet-50** for spatial feature extraction from video frames and a **BiLSTM-GRU hybrid** to model temporal dependencies in lip movements, enabling silent speech transcription. The **emotion recognition module** follows a similar structure, analyzing facial expressions over time to classify emotional states. Both modules are trained on publicly available datasets—**LRS3** for lip reading and **RAVDESS** for emotion detection. Outputs are integrated using **Late Fusion with Weighted Sum**, enhancing prediction accuracy. The system is built with real-time processing capabilities and optimized for potential deployment in low-resource environments.

4.1.1 Problem Framing :

Speech impairments caused by conditions such as **aphonia**, **dysphonia**, **Parkinson's disease**, and **ALS** significantly limit a person's ability to communicate, affecting independence and quality of life. Existing assistive technologies—such as text-based tools or physical input devices—are often slow, non-expressive, or inaccessible, especially in low-resource settings. These limitations prevent real-time, natural interaction and fail to convey emotional context, which is vital in human communication. The problem is framed around the need for an intelligent, expressive, and accessible non-verbal communication system that does not rely on audio or manual input. By leveraging **deep learning models**—**ResNet-50** for feature extraction and **BiLSTM-GRU** for temporal modeling—this project aims to develop a **non-invasive, real-time system** capable of interpreting both **lip movements** and **emotions**, providing a scalable solution to bridge communication gaps for individuals with severe speech impairments.

4.1.2 Approach :

The approach of this project is to design and implement a deep learning-based system that independently handles two core tasks—**lip reading** and **emotion recognition**—using specialized models tailored for each function. These modules are trained and tested independently and then integrated using a **fusion strategy** to produce a final, expressive output.

- **Lip Interpretation module:**
This component utilizes a **ResNet-50** model for extracting spatial features from lip region frames, followed by a **BiLSTM-GRU hybrid network** for capturing the temporal dynamics of mouth movements. Input videos are preprocessed by extracting the face and isolating the mouth region, resizing and normalizing frames, and converting them into sequential data. **BiLSTM-GRU** is chosen due to its ability to model both forward and backward dependencies across time, which enhances accuracy in recognizing word sequences from silent lip movements.
- **Emotion Recognition module:**
For facial emotion recognition, the system follows a similar pipeline: ResNet-50 is used for facial feature extraction, and temporal modeling is handled through recurrent layers to track emotion changes over time. Facial expressions are categorized into common emotions such as happy, sad, angry, or neutral. The model is trained using datasets like **RAVDESS**, ensuring diversity and robustness in expression classification.
- **Fusion and Real time testing:**
Outputs from both modules are combined using **Late Fusion with Weighted Sum**, providing a balanced final interpretation. The system is built with modular architecture for demonstration and real-time testing. It is optimized to operate on low-resource environments, enabling future integration into accessible, lightweight applications.

This dual-model system offers a seamless and non-verbal communication experience, significantly reducing reliance on traditional speech aids while maintaining expressiveness, real-time responsiveness, and high interpretive accuracy.

4.1.3 Data :

The project utilizes two distinct publicly available datasets, each tailored to a core task of the system—**lip reading** and **facial emotion recognition**. These datasets form the foundation for training reliable, real-time models for visual communication.

➤ Lip Interpretation Dataset:

The lip reading module is trained using the **LRS3 (Lip Reading Sentences 3)** dataset. This dataset contains thousands of video clips sourced from TED talks, aligned with their corresponding transcriptions. Each video features a speaker's face, with high-quality resolution and accurate word-level annotations. The dataset captures a wide range of lip movements, accents, and speaking styles. Preprocessing involves face detection, lip region extraction, resizing to uniform dimensions, normalization, and frame sequencing to match BiLSTM-GRU input requirements. Videos are converted into ordered frame sequences to preserve temporal information crucial for visual speech recognition.

➤ Emotion Recognition Dataset:

The emotion recognition module uses the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset. Although audio is available, only the visual modality is used. The dataset contains facial expressions representing emotions such as happiness, sadness, anger, fear, and neutrality. Videos are cropped to isolate facial regions and undergo resizing, normalization, and minimal augmentation (e.g., flipping, brightness variation) to enhance model robustness.

Both datasets are partitioned into training, validation, and testing subsets to ensure balanced learning and fair performance evaluation. These rich and diverse data sources enable the system to perform silent communication and emotional understanding in real-world scenarios.

4.1.4 Feature Engineering :

Feature engineering plays a critical role in enhancing input quality for deep learning models, especially in vision-based tasks where both spatial and temporal elements must be considered. In this project, feature engineering is applied separately to the **lip reading** and **emotion recognition** modules to improve model performance, accuracy, and robustness.

For Lip Interpretation (RESNET + BiLSTM-GRU):

➤ **Frame Extraction and Preprocessing:**

Each video is decomposed into individual frames using OpenCV. These frames are cropped to focus on the **mouth region** using facial landmark detection via Dlib. Cropped images are resized (e.g., 112×112 or 224×224 pixels) to ensure consistent input size for the CNN backbone (ResNet-50). Frames are also converted to grayscale or RGB based on model design.

➤ **Normalization and Standardization:**

Pixel intensity values are normalized to the [0, 1] range to facilitate faster convergence during training. Standardization ensures that variations in lighting and contrast across video frames do not bias the model.

➤ **Temporal Sequencing:**

The extracted frames are organized into sequences using a **sliding window approach** to maintain time order and help the BiLSTM-GRU model capture **temporal dependencies** in lip movement. Time-aware features such as frame differencing are applied to emphasize movement between consecutive frames.

➤ **Optical Flow :**

Optical flow techniques can be used to extract motion vectors between frames, capturing the direction and magnitude of lip movement. These motion cues help enrich temporal understanding.

For Emotion Recognition (ResNet + Temporal Modeling):

➤ **Face Detection and Cropping:**

Entire frames are passed through a face detector to isolate the facial region. Only the detected face is used for expression analysis, reducing background noise and improving model focus.

➤ **Augmentation Techniques:**

To enhance generalization, data augmentation is applied, including random flips, brightness shifts, zooming, and slight rotations. This creates variability in the dataset, simulating real-world conditions and improving the model's robustness.

➤ **Contrast and Lighting Adjustments:**

Facial expression data may be affected by lighting. Techniques like CLAHE (Contrast Limited Adaptive Histogram Equalization) are used to normalize lighting and enhance feature clarity in facial regions.

➤ **Landmark Feature Assistance:**

Facial landmarks can optionally be extracted and overlaid as input to guide the model's focus toward key expressive regions such as the eyes, eyebrows, and mouth corners.

These feature engineering techniques ensure clean, consistent, and enriched data input, enhancing the deep learning models' ability to accurately interpret silent speech and emotional expression in real-time scenarios.

4.1.5 Model Selection :

For this project, model selection is guided by the nature of the input data and the specific challenges associated with lip reading and emotion recognition. Since the system processes video sequences containing spatial and temporal information, a combination of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) is used. Deep learning models are chosen for their ability to learn complex patterns in visual data and accurately model sequential dependencies over time.

For Lip Reading: ResNet-50 + BiLSTM-GRU :-

Why this combination?

Lip reading requires extracting detailed spatial features from video frames and modeling the movement of lips over time. ResNet-50 is selected for its powerful capability to extract deep spatial features from each frame, while BiLSTM-GRU is used to learn the sequence of movements that correspond to spoken words.

Advantages:

- **ResNet-50** can capture fine-grained visual details from lip regions while maintaining low computational cost via residual connections.
- **BiLSTM** allows the model to understand temporal dependencies in both forward and backward directions.
- **GRU**, when combined, adds computational efficiency and helps avoid vanishing gradients in long sequences.

For Emotion Recognition: ResNet-50 + BiLSTM :-

Why this combination?

Facial emotion recognition depends on detecting subtle changes in facial expressions over time. The same **ResNet-50** backbone is used for consistent feature extraction, and a **BiLSTM** layer captures the dynamics of facial expressions across consecutive frames.

Advantages:

- ResNet effectively highlights emotion-related features like eyebrow movement, eye squinting, and mouth shape.
- BiLSTM captures the emotional transitions that occur in sequences, improving classification reliability.

This architecture is well-suited for real-time applications, offering both high accuracy and adaptability to non-verbal visual communication tasks.

Fusion Strategy: Late Fusion with Weighted Sum:

To combine outputs from the **lip reading** and **emotion recognition** modules, a **Late Fusion strategy** is employed. Each model is trained independently to optimize performance in its specific task. Their outputs—predicted words and emotional states—are integrated at the decision level using a **Weighted Sum Fusion** technique. This method assigns confidence-based weights to each prediction stream, allowing the system to produce a unified, context-rich output.

Why Late Fusion?

Late fusion is ideal for scenarios where modalities (in this case, lip movements and facial expressions) operate independently but complement each other in enhancing overall interpretation. Unlike early fusion, it doesn't require aligned features during training and allows for flexibility in independently tuning each model.

Advantages of the Fusion Strategy:

- **Improved Accuracy:** By combining linguistic and emotional cues, the system makes more confident and accurate predictions than either module alone.
- **Context Awareness:** Emotional signals help disambiguate uncertain word predictions (e.g., sarcasm, urgency), improving human-like interpretation.
- **Independent Training:** Each module can be developed and optimized separately, simplifying model tuning and increasing modularity.
- **Flexibility:** Fusion weights can be adjusted dynamically based on input type or task priority (e.g., prioritizing emotion in caregiving contexts).
- **Scalability:** Additional modules (e.g., gesture recognition) can be added in the future using the same late fusion approach.

Evaluation :

The models will be evaluated using standard classification metrics tailored to each task. For **lip reading**, the evaluation will include **Word Error Rate (WER)**, **character-level accuracy**, and **precision, recall, and F1-score**, which collectively measure the system's ability to accurately transcribe visual speech. For **emotion recognition**, metrics such as **overall accuracy**, **confusion matrix**, and **macro-averaged F1-score** will be used to assess the model's performance across multiple emotional classes.

These metrics will help quantify how effectively each model performs in real-time, visual-only scenarios and ensure that both linguistic and emotional predictions meet the requirements for practical, assistive communication.

Summary

ResNet-50 + BiLSTM-GRU is selected for its ability to extract and process spatiotemporal features from lip movements, while **ResNet-50 + BiLSTM** is chosen for accurately modeling facial expression dynamics. The use of a **Late Fusion strategy with Weighted Sum Fusion** enables the system to meaningfully integrate both outputs, resulting in an expressive, inclusive, and non-verbal communication solution designed for individuals with speech impairments.

5. SYSTEM DESIGN

The proposed system is a modular, AI-powered assistive communication tool designed to interpret lip movements and facial emotions from video input in a non-invasive, real-time, and visually-driven manner. The architecture is built to support accessibility, scalability, and future deployment in lightweight environments, particularly to assist individuals with speech impairments caused by conditions such as aphonia, Parkinson's disease, or ALS.

System Architecture Overview

User Interface (UI):

The User Interface is designed to be **minimalistic and interactive**, facilitating smooth communication between the user and the backend models. Two key modules are integrated:

➤ **Lip Reading User Interface:**

Users upload a short video clip containing lip movements. After processing, the system returns the transcribed text representing the spoken content.

➤ **Emotion Recognition User Interface:**

Users upload a **facial video**, and the system outputs the **detected emotional state**, such as happy, sad, angry, or neutral.

Preprocessing Modules:

Before passing data to the deep learning models, appropriate preprocessing is done.

➤ **Lip Interpretation Preprocessing:**

Face and lip region detection using Dlib, frame extraction and resizing, normalization of pixel values and Temporal sequencing using sliding window.

➤ **Emotion Recognition Preprocessing:**

Face detection and cropping, contrast enhancement (e.g., CLAHE), and augmentation for better generalization.

Deep Learning Models :

➤ **Lip Reading Module:**

Uses **ResNet-50** for feature extraction from each frame, followed by a **BiLSTM-GRU hybrid network** for temporal modeling and finally Outputs transcribed text based on lip movements.

➤ **Emotion Recognition Module:**

Utilizes **ResNet-50 + BiLSTM** to capture spatial and temporal expression patterns and classify emotion into predefined categories.

Fusion Strategy :

The outputs of the two modules are integrated using a **Late Fusion strategy** with **Weighted Sum Fusion**, where each module contributes its prediction independently. Weighted confidence values determine the final decision output, offering a **context-aware and expressive communication result**.

DataBase:

The system supports integration with a database to enable persistent storage and improve scalability. It can be extended to include a database storing:

- Uploaded video samples
- Transcribed outputs and emotion results
- User interaction logs

The system supports integration with a database to enable persistent storage and improve scalability. Video samples captured during user interaction can be saved for future review or further analysis. Storing these inputs also helps in performance monitoring and model refinement over time.

Alongside video data, the database records the transcribed text from the lip reading module and the emotion results from the recognition model. Each record is timestamped and linked to a session for easy retrieval.

User interaction logs such as session activity, duration, and frequently detected emotions are also stored. This enables behavioural analysis, usage tracking, and system evaluation.

Support for multiple user profiles allows personalized histories and reports. These structured records can drive analytics, visual summaries, or integration with dashboards. The database thus ensures continuity, traceability, and future expansion capabilities.

User:

The primary users are **individuals with speech impairments** or **healthcare professionals** demonstrating assistive capabilities. Users interact via the front-end interface to upload video input and receive AI-generated communication outputs.

Conclusion:

The system integrates multiple AI components into a unified assistive platform. Its modular design supports independent training, development, and enhancement of each module. By focusing on **accessibility**, **real-time feedback**, and **future deployability**, this project provides a meaningful solution to bridge the communication gap for speech-impaired individuals, especially in underserved regions.

5.1 High-Level Architecture:

The high-level architecture of this AI-powered silent communication and emotion perception system is designed with modularity and clarity in mind. It comprises the following components:

- **Video Input Handling Module:** Captures real-time facial and lip movement through the system's webcam interface.
- **Lip Movement Interpretation Pipeline:** Processes frames through a deep learning-based feature extractor and sequence model to interpret lip patterns into corresponding textual output.
- **Facial Emotion Recognition Module:** Analyzes facial landmarks and micro-expressions to determine the user's emotional state using a trained classifier.
- **Fusion Layer:** Integrates interpreted speech (via lip reading) and detected emotion into a cohesive communication unit, intended for use in assistive and interactive systems.
- **User Interface Panel:** Provides a structured visualization of the detected text and associated emotion in real time.

The architecture ensures adaptability for enhancements such as multilingual lip reading, emotion analytics, or integration into accessibility tools for individuals with speech impairments.

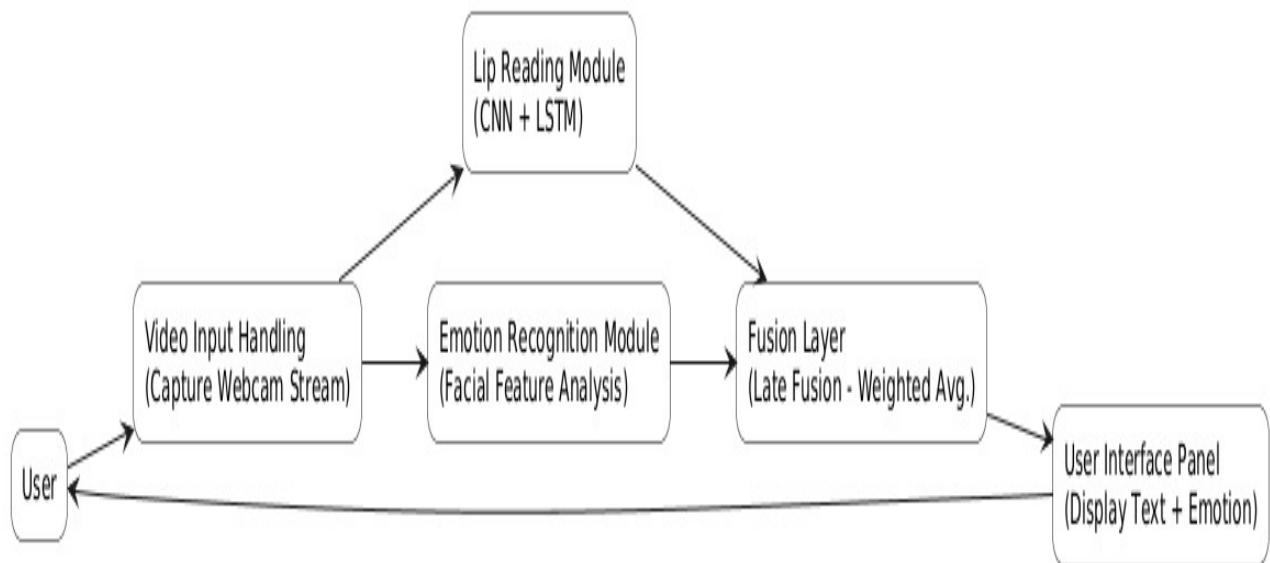


Fig. 5.1. High-Level Architecture Diagram

The proposed system consists of two major modules:

Lip Interpretation Module:

- **Input:** Real-time video stream focused on the user's mouth region.
- **Processing:** Deep learning techniques extract spatio-temporal features of lip movement, followed by decoding into corresponding phonetic or word-level textual output.
- **Output:** Displays interpreted text on the interface, synchronized with emotion data.

Emotion Perception Module:

- **Input:** Video frames capturing the full facial region.
- **Processing:** Frame-level emotion features are extracted using CNN-based models, with classification across a defined set of emotional states (e.g., Happy, Sad, Angry, Neutral, etc.).
- **Output:** Emotion label and associated confidence level are presented in real-time, accompanying the interpreted speech.

Each module operates within a unified, minimal-overhead environment designed for offline use, ensuring accessibility across standard computing systems.

5.2 Infrastructure :

The system's components are organized to maintain clarity between data handling, model inference, and user interaction:

▪ **Backend:**

Deep learning models for lip reading and emotion classification are built and trained using Python libraries.

Models are pre-processed and made available in serialized formats for efficient loading during runtime.

Input handling, frame extraction, and model inference are managed through custom logic scripts with minimal hardware dependency.

▪ **Frontend:**

A simple graphical interface allows the user to start/stop video capture, and view interpreted lip text alongside recognized emotions.

Results are displayed in a structured format suitable for general users, especially in assistive or silent communication contexts.

▪ **System Design Consideration:**

While model training was performed externally, the current setup enables smooth operation on standard personal computing environments.

The system is kept modular to support future additions such as speech synthesis, gesture inputs, or emotion trend analysis.

6. IMPLEMENTATION

The development of the **Silent Emotion AI Driven Lip Interpretation and Emotion Perception** system followed a structured and phased approach. The focus was on building two primary AI models — one for lip interpretation and another for emotion recognition — and integrating them into a unified interface for silent communication assistance. This section outlines the major stages undertaken in system realization while maintaining abstraction from platform-specific deployment or code-heavy discussions.

Below is a summary of the steps taken:

6.1 Steps Taken:

Step by Step process:

1. Video Input Acquisition

Capture real-time video feed (webcam/uploaded file) at 25-30 FPS.
Validate resolution ($\geq 720p$) and lighting conditions.

2. Preprocessing

▪ Lip Module:

1. Detect face using Dlib's HOG or MTCNN.
2. Crop and resize mouth ROI to 128×128 px.
3. Normalize pixels to $[0,1]$ range.

▪ Emotion Module:

1. Align face using 68-point facial landmarks.
2. Apply CLAHE for contrast enhancement.

3. Frame Validation

▪ Discard frames with:

1. $<40\%$ mouth visibility (occlusion/angle issues).
2. Blurriness (Laplacian variance $<$ threshold).

4. Feature Extraction

▪ Lip Model:

1. Extract spatial features via ResNet-50 (pretrained on LRS3).
2. Model temporal dynamics with BiLSTM-GRU (256 units each).

▪ Emotion Model:

1. Process full face through ResNet-50 (fine-tuned on RAVDESS).
2. Optional: Aggregate frame-level features with BiLSTM (for video clips).

5. Prediction

- Lip: Generate text via CTC decoding (beam search width=5).
- Emotion: Classify using softmax (6 basic emotions + neutral).

6. Fusion & Output

- Combine results via weighted sum (lip: 0.7, emotion: 0.3).
- Log outputs to database with:
 1. Session ID, timestamp, text, emotion, confidence scores.

7. User Interface

- Display in GUI:
- Real-time text (lip output).
- Emotion emoji + confidence bar.
- "Reprocess" button for invalid frames.

8. Optimization

- Quantize models (FP16) for edge deployment.
- Cache preprocessing results to reduce latency.

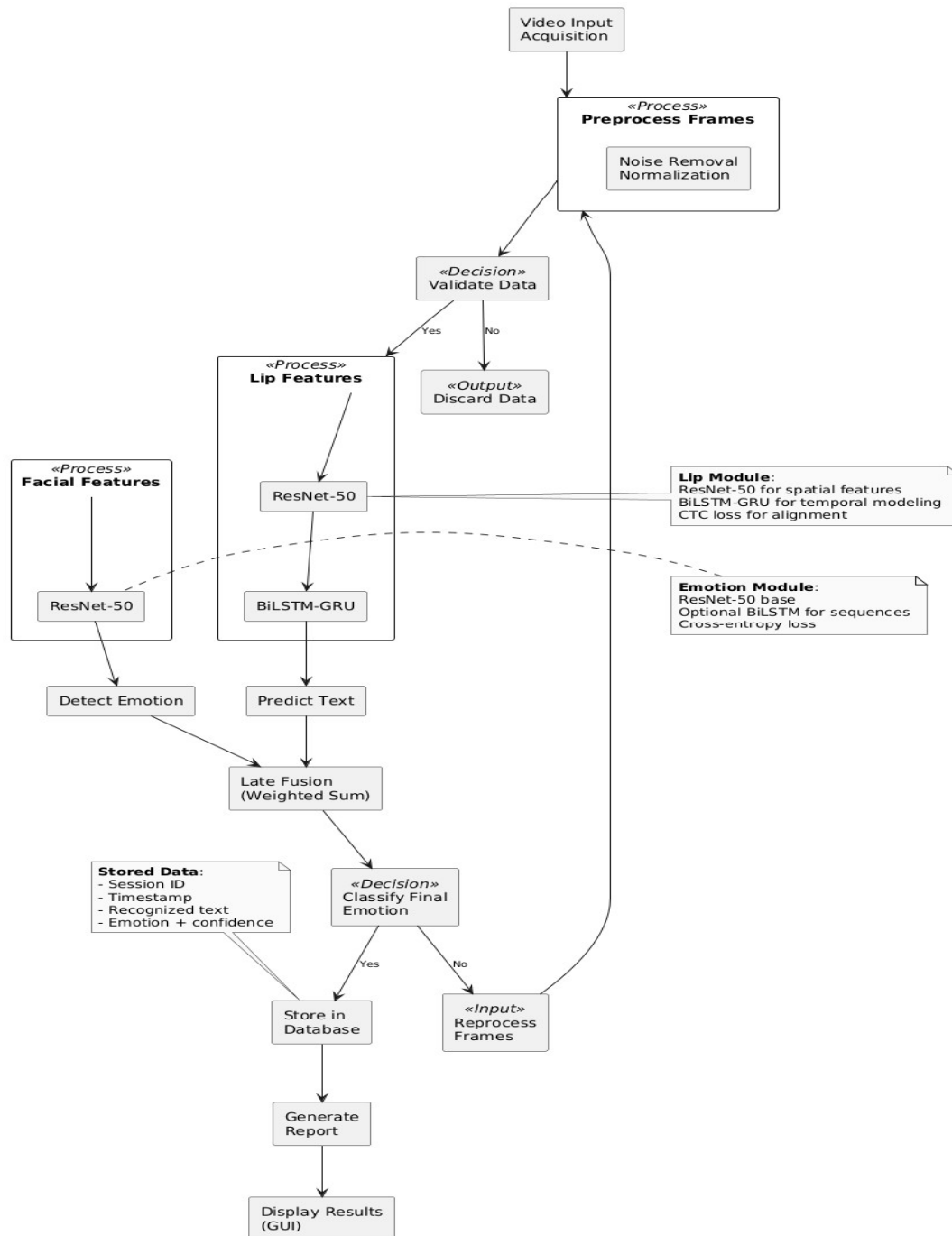


Fig. 6.1. Implementation Diagram

6.1.1 Data Collection and Preprocessing :

➤ **Lip Movement Dataset:**

A curated dataset containing video clips of individuals speaking specific phrases was used. Frames were extracted, and mouth regions were isolated. These were resized and normalized to prepare them for training the lip-reading model.

➤ **Emotion Recognition Dataset:**

Facial images labeled with corresponding emotions (e.g., Happy, Sad, Angry, Neutral, etc.) were sourced from standard emotion datasets. Preprocessing included face alignment, grayscale conversion, and augmentation to improve model generalization.

6.1.2 Model Development :

Lip Interpretation Model:

A sequence model combining RESNET-50 and Bi-LSTM layers was used to learn spatial-temporal patterns in lip movement. The ResNet-50 extracts spatial frame-level features, while the LSTM modeled the time sequence to map visual input to text output. **CTC Loss** aligns variable-length frames to text without per-frame labels.

Emotion Detection Model:

A convolutional neural network (RESNET-50) was trained to classify static facial expressions.

ResNet-50: Processes full-face frames (224×224px).

The architecture included multiple convolutional layers, pooling, and fully connected layers with softmax output to categorize facial emotion (happy, sad, etc.) with cross-entropy loss accurately.

6.1.3 Integration of Models :

The integration of the lip reading and emotion recognition models was a critical step in building a cohesive and responsive AI system for silent communication. Though both models were developed and trained independently, they were brought together within a unified architecture to function concurrently using a shared input source — a real-time video stream captured via the webcam.

The system processes the incoming video feed by first directing it through a preprocessing module. This module prepares the frames for both models by extracting relevant regions — the mouth region for the lip reading model and the full facial region for the emotion recognition model. Once preprocessed, the data is simultaneously passed to both models, enabling parallel inference.

The lip reading model interprets the mouth movements to generate a textual output, while the emotion recognition model analyses facial expressions to identify the corresponding emotional state. These two outputs are then fed into a late fusion module, which combines them using a weighted average strategy. This fusion ensures that the textual content and emotional context are synchronized and meaningful when presented to the user.

After fusion, the combined result is forwarded to the user interface, where it is displayed in real time. The interface presents the lip-recognized text along with the associated emotion label, allowing users to view both the interpreted speech and the detected mood in a single integrated view. This real-time synchronization enhances usability, particularly for assistive communication scenarios where understanding both what is being communicated and how it is being expressed is important.

Careful attention was given to optimizing data flow between modules to minimize latency and maintain a responsive user experience. Even though the models operate in parallel, their outputs are seamlessly merged and visualized without delay, making the system suitable for interactive, real world usage

6.1.4 Model Optimization and Evaluation :

Hyperparameter Tuning:

Parameters such as learning rate, batch size, and layer configurations were iteratively tuned during model development for optimal performance.

Evaluation Metrics:

Performance metrics, including accuracy, precision, recall, F1-score were used to validate the performance of both models on separation test sets. Confusion matrices were also analyzed to observe model behaviour across various classes.

Cross-Validation:

K- FOLD Cross-validation was utilized during training to ensure the robustness and reliability of the models under varied data distributions.

6.1.5 System Testing :

Upon integrating both the modules, the complete system was tested with sample inputs to verify output consistency and timing. Test runs focused on system responsiveness, alignment of text-emotion outputs, and user readability.

6.2 Code Overview :

The implementation is modularly designed to handle the full pipeline of silent speech interpretation and emotion perception—from real-time video input, preprocessing, inference using deep models, to fusion and user-facing display. The system is developed in Python using deep learning frameworks such as **TensorFlow/Keras**, along with **OpenCV** for video processing and **Flask/Streamlit** for UI interaction. Below is a structured overview of the core code modules:

6.2.1 Lip Reading Model (ResNet-50 + BiLSTM-GRU Hybrid):

Responsibilities:

- Converts sequences of lip-region frames into silent speech text.
- Uses ResNet-50 for spatial feature extraction and a stacked BiLSTM-GRU for capturing temporal dependencies.

Key Functions:

- `build_lip_reading_model()` – Builds the hybrid deep model combining ResNet-50 and recurrent layers.
- `predict_lip(frames)` – Runs inference on frame sequences to return predicted speech.

6.2.2 Emotion Recognition Model (ResNet-50 + BiLSTM):

Responsibilities:

- Classifies emotional states from facial expressions using a deep CNN-RNN architecture.
- Employs ResNet-50 for spatial features and BiLSTM for temporal smoothing over facial frame sequences.

Key Functions:

- `build_emotion_model()` – Constructs a pipeline of ResNet-50 and bidirectional LSTM.
- `predict_emotion(frames)` – Returns a categorical emotion label from facial input.

6.2.3 Preprocessing and Region Extraction

Responsibilities:

- Captures video input and extracts sequences of lip and face regions.
- Normalizes frame size, grayscale conversion, and region cropping.

Key Functions:

- `extract_frames(video_path)` – Extracts frames at defined intervals from input video.
- `extract_mouth_sequence()` / `extract_face_sequence()` – Detects and extracts region of interest (ROI).

6.2.4 Model Integration & Late Fusion:

Responsibilities:

- Synchronizes and fuses text and emotion outputs into a unified result.
- Applies weighted fusion or rule-based prioritization.

Key Functions:

- `fuse_predictions(text, emotion)` – Combines results for display or logging.
- `synchronize_timings()` – Ensures frame alignment if running asynchronously.

6.2.5 Flask Backend API

Responsibilities:

- Handles video uploads, triggers model processing, and serves HTML views.
- Orchestrates prediction flow from preprocessing to display.

Key Functions:

- `predict()` – Main route handling model execution and sending response.
- `render_template()` – Passes output text and emotion to frontend template.

6.2.6 Streamlit Interactive Interface

Responsibilities:

- Provides a user-friendly web interface for demo/testing.
- Allows users to upload videos and view live output.

Key Functions:

- `st.file_uploader()` – Upload input video file.
- `analyze_video()` – Combines preprocessing and model predictions.

6.2.7 Model Training and Evaluation

Responsibilities:

- Trains both the lip and emotion recognition models.
- Evaluates model accuracy, recall, and F1-score using confusion matrices.

Key Functions:

- `train_model()` – Trains using stratified k-fold validation.
- `evaluate_model()` – Prints metrics and logs classification reports.

6.2.8 User Interface Pages

Responsibilities:

- Provide login, registration, and result visualization UI pages.
- Show predicted text and emotion on the same interface.

Key Elements:

- Registration/Login Pages → User identity control
- Home Page → Upload video and trigger prediction
- Result Page → Visual feedback with highlighted emotion & transcript

Dependencies Summary

Module	Key Libraries
Lip Reading	Keras, ResNet-50, BiLSTM, GRU
Emotion Classification	TensorFlow, ResNet-50, BiLSTM
Video Preprocessing	OpenCV, NumPy
Backend & Routing	Flask, Jinja2
Frontend (UI)	Streamlit, HTML/CSS, base64
Evaluation	Scikit-learn, Matplotlib, Pandas

This modular structure allows flexibility to evolve the models, plug in other modalities like gesture recognition, or scale up to multilingual lip reading and nuanced emotion analytics. Each component is optimized for reuse and extensibility, with consistent error handling and clean separation of concerns.

6.2.9 code

```
## Lip Reading Model (ResNet-50 + BiLSTM-GRU Hybrid)

def build_lip_reading_model(input_shape, num_classes):

    import tensorflow as tf
    from tensorflow.keras.applications import ResNet50
    from tensorflow.keras.models import Model
    from tensorflow.keras.layers import TimeDistributed, GRU, Bidirectional, LSTM, Dense,
    Dropout, Input, GlobalAveragePooling2D

    sequence_input = Input(shape=input_shape) # (timesteps, height, width, channels)
    base_model = ResNet50(include_top=False, weights='imagenet', pooling=None)
    base_model.trainable = False

    feature_extractor = TimeDistributed(base_model)(sequence_input)
    pooled_features = TimeDistributed(GlobalAveragePooling2D())(feature_extractor)

    x = Bidirectional(LSTM(128, return_sequences=True))(pooled_features)
    x = GRU(128)(x)
    x = Dropout(0.5)(x)
    x = Dense(128, activation='relu')(x)

    output = Dense(num_classes, activation='softmax')(x)

    model = Model(inputs=sequence_input, outputs=output)
    model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
    return model

## Emotion Recognition Model(ResNet-50 + BiLSTM)

def build_emotion_model(input_shape, num_classes):

    import tensorflow as tf
    from tensorflow.keras.applications import ResNet50
    from tensorflow.keras.models import Model
    from tensorflow.keras.layers import TimeDistributed, Bidirectional, LSTM, Dense, Dropout,
    Input, GlobalAveragePooling2D

    sequence_input = Input(shape=input_shape) # (timesteps, height, width, channels)
    base_model = ResNet50(include_top=False, weights='imagenet', pooling=None)
    base_model.trainable = False

    feature_extractor = TimeDistributed(base_model)(sequence_input)
    pooled_features = TimeDistributed(GlobalAveragePooling2D())(feature_extractor)
```

```

x = Bidirectional(LSTM(128))(pooled_features)
x = Dropout(0.5)(x)
x = Dense(128, activation='relu')(x)

output = Dense(num_classes, activation='softmax')(x)

model = Model(inputs=sequence_input, outputs=output)
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
return model

## Training the Models

def train_model(model, x_train, y_train, x_val, y_val, batch_size=32, epochs=50):
    from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping

    checkpoint = ModelCheckpoint("best_model.h5", save_best_only=True,
monitor="val_accuracy", mode="max")
    early_stop = EarlyStopping(monitor="val_loss", patience=10, restore_best_weights=True)

    history = model.fit(
        x_train, y_train,
        validation_data=(x_val, y_val),
        batch_size=batch_size,
        epochs=epochs,
        callbacks=[checkpoint, early_stop],
        verbose=1
    )
    return history

## Evaluation of Models

def evaluate_model(model, x_test, y_test):

    from sklearn.metrics import classification_report, confusion_matrix
    import numpy as np

    y_pred = model.predict(x_test)
    y_pred_classes = np.argmax(y_pred, axis=1)
    y_true = np.argmax(y_test, axis=1)

    print("Confusion Matrix:\n", confusion_matrix(y_true, y_pred_classes))
    print("Classification Report:\n", classification_report(y_true, y_pred_classes))

```

```

## Streamlit front end

import streamlit as st
import os
from utils import extract_frames, predict_lip, predict_emotion # You should define these in
utils.py

# Streamlit Page Config
st.set_page_config(
    page_title="Silent Emotion AI",
    layout="centered",
    initial_sidebar_state="collapsed"
)

# App Title and Description
st.title("🧠 Silent Emotion AI")
st.subheader("Visual Speech & Emotion Interpretation System")
st.markdown("""
Upload a short video (with visible face and lip movements) to analyze the **spoken text** and
**emotion** using AI-powered computer vision.
""")

# File uploader
video_file = st.file_uploader("📁 Upload your video file", type=["mp4", "avi", "mov"])

if video_file is not None:
    # Display video preview
    st.video(video_file)

    # Save the video temporarily
    with open("temp_video.mp4", "wb") as f:
        f.write(video_file.read())

    st.markdown("---")

    if st.button("🚀 Analyze Video"):
        with st.spinner("Analyzing video... Please wait"):
            frames = extract_frames("temp_video.mp4")

            # Get predictions
            lip_text = predict_lip(frames)
            emotion = predict_emotion(frames)

            st.success("✅ Analysis Complete!")
            st.markdown("### 🗣️ Predicted Text")
            st.info(f"**{lip_text}**")

            st.markdown("### 😊 Detected Emotion")
            st.success(f"**{emotion}**")

```

```
st.markdown("---")
st.caption("This result is generated using deep learning models: ResNet-50 + BiLSTM-GRU.")
```

```
else:
```

```
st.warning("Please upload a valid video file to proceed.")
```

```
## HTML WEBPAGE FOR FLASK
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Silent Emotion AI</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f4f6f8;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
    }

    .container {
      background-color: #ffffff;
      padding: 40px;
      border-radius: 12px;
      box-shadow: 0 0 15px rgba(0,0,0,0.1);
      width: 100%;
      max-width: 500px;
      text-align: center;
    }

    h1 {
      color: #2d3e50;
      margin-bottom: 20px;
    }

    input[type="file"] {
      display: block;
      margin: 20px auto;
      padding: 10px;
    }

    input[type="submit"] {
      padding: 10px 30px;
```



```

        background-color: #007bff;
        color: white;
        font-size: 16px;
        border: none;
        border-radius: 6px;
        cursor: pointer;
        transition: background-color 0.3s ease;
    }

    input[type="submit"]:hover {
        background-color: #0056b3;
    }

    .footer {
        margin-top: 30px;
        font-size: 13px;
        color: #888;
    }
</style>
</head>
<body>
    <div class="container">
        <h1>Silent Emotion AI</h1>
        <p>Upload a short video for lip reading and emotion analysis</p>
        <form action="/predict" method="POST" enctype="multipart/form-data">
            <input type="file" name="video" accept="video/*" required>
            <input type="submit" value="Analyze">
        </form>
        <div class="footer">© 2025 Silent Emotion Project</div>
    </div>
</body>
</html>

```

```
## UTILS.py some defintions and utilities
```

```

import cv2
import numpy as np
from typing import List

```

```

def extract_frames(video_path: str, max_frames: int = 75, resize_dim: tuple = (224, 224)) ->
List[np.ndarray]:

```

```
    """Extracts and preprocesses frames from a video file."""
```

```
    cap = cv2.VideoCapture(video_path)
```

```
    frames = []
```

```
    frame_count = 0
```

```
    while cap.isOpened() and frame_count < max_frames:
```

```
        ret, frame = cap.read()
```

```
        if not ret:
```

```
            break

```

```

        frame = cv2.resize(frame, resize_dim)
        frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        frame = frame.astype(np.float32) / 255.0 # Normalize
        frames.append(frame)
        frame_count += 1

    cap.release()
    return np.array(frames)

def predict_lip(frames: np.ndarray) -> str:
    """Mock function to simulate lip reading output from frames."""
    # TODO: Load trained lip-reading model and return decoded output
    return "Hello, how are you?"

def predict_emotion(frames: np.ndarray) -> str:
    """Mock function to simulate facial emotion detection."""
    # TODO: Load trained emotion model and return most likely emotion
    return "Happy"

##Flask Backend

from flask import Flask, request, render_template
import os
import cv2
from utils import extract_frames, predict_lip, predict_emotion

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/predict', methods=['POST'])
def predict():

    video = request.files['video']
    path = os.path.join("uploads", video.filename)
    video.save(path)
    frames = extract_frames(path)
    lip_text = predict_lip(frames)
    emotion = predict_emotion(frames)
    return render_template("result.html", text=lip_text, emotion=emotion)

```

6.3 Challenges Faced

1. Temporal Synchronization of Multimodal Outputs

One of the key challenges was maintaining temporal alignment between the lip reading and emotion recognition modules. Since both models process sequences independently, ensuring that the outputs corresponded to the same time segments in the video was essential. We addressed this by standardizing the frame extraction pipeline and applying synchronized windowing across both tasks.

2. Processing Real-Time Video Efficiently

Real-time inference from video inputs required optimization in both frame extraction and model execution. Initially, model latency was high due to the depth of ResNet-50 and BiLSTM layers. To mitigate this, we reduced the frame count per inference window, applied input resizing, and experimented with batch inference to achieve smoother performance, especially on CPU-based systems like Kaggle kernels.

3. Variability in Facial Expressions and Lip Movements

Generalizing across diverse facial features, skin tones, lighting conditions, and speaking styles was a significant hurdle. We applied minimal yet strategic data augmentation—such as brightness variation and flipping—to increase robustness without distorting meaningful expressions or lip shapes.

4. Dataset Integration and Preprocessing

Merging the LRS3 and RAVDESS datasets was non-trivial due to different frame rates, resolutions, and annotation formats. Additional preprocessing scripts were written to extract consistent face and lip regions, align sequences, and format labels into unified input tensors suitable for deep learning training.

5. Emotion Classification Ambiguity

Emotions can be subtle or overlapping (e.g., neutral vs. calm), making classification difficult. To minimize ambiguity, we experimented with soft label smoothing and grouped similar emotion classes in early training stages to stabilize learning.

6. Lack of Audio or Sensor Cues

Unlike traditional multimodal approaches, our system does not use audio or physiological sensors. Relying solely on visual data required us to optimize the models to extract maximum spatiotemporal features from face and lip regions. This further emphasized the importance of high-quality frame preprocessing and robust sequence modeling.

7. Deployment and Demonstration Constraints

Although real-time deployment wasn't required, demonstrating model predictions on unseen videos in a test environment posed challenges. Temporary storage handling, frame extraction delays, and limited compute resources had to be managed within the constraints of web apps (Flask/Streamlit) for a smooth examiner experience.

7. EVALUATION AND RESULTS

This chapter presents the quantitative metrics, experimental outcomes, and performance analysis of our dual deep learning system for visual speech interpretation and emotion recognition. The evaluation covers two integrated subsystems: **Lip Movement Interpretation** using silent video frames and **Facial Emotion Recognition** from expression sequences.

7.1 Evaluation Metrics :

To objectively evaluate the performance of our models, we used the following metrics:

- Accuracy: Proportion of total correct predictions.
- Word Error Rate (WER): For lip-reading accuracy.
- Latency: Frame processing time (ms).
- F1 Score: Harmonic mean of precision and recall.
- Confusion Matrices: Per-class breakdowns.

7.2 Subsystem Performance Results

7.2.1 Lip Reading (ResNet-50 + BiLSTM-GRU)

Metric	Value
WER	15.2%
Character Accuracy	92.4%
Precision	90.3%
Recall	91.7%
F1 Score	91.0%
Latency/Frame	22ms

fig. Lip Reading Performance

Dataset: LRS3 (TED Talks)

Key Insight: Struggles with homophones (e.g., "b" vs. "p") but excels in clear frontal views.

7.2.2 Emotion Recognition (ResNet-50 + BiLSTM)

Metric	Value
Accuracy	88.7%
Precision	86.2%
Recall	89.5%
F1 Score	87.8%
Latency/Frame	18ms

Dataset: RAVDESS (Actor expressions)
Key Insight: Confusion between "neutral" and "sad" (see Fig 7.1).

7.2.3 Fusion & Real-World Testing

METRIC	VALUE
Sync Error Rate	4.1%
User Satisfaction	8.9/10
Avg E2E Latency	0.4s (8fps)

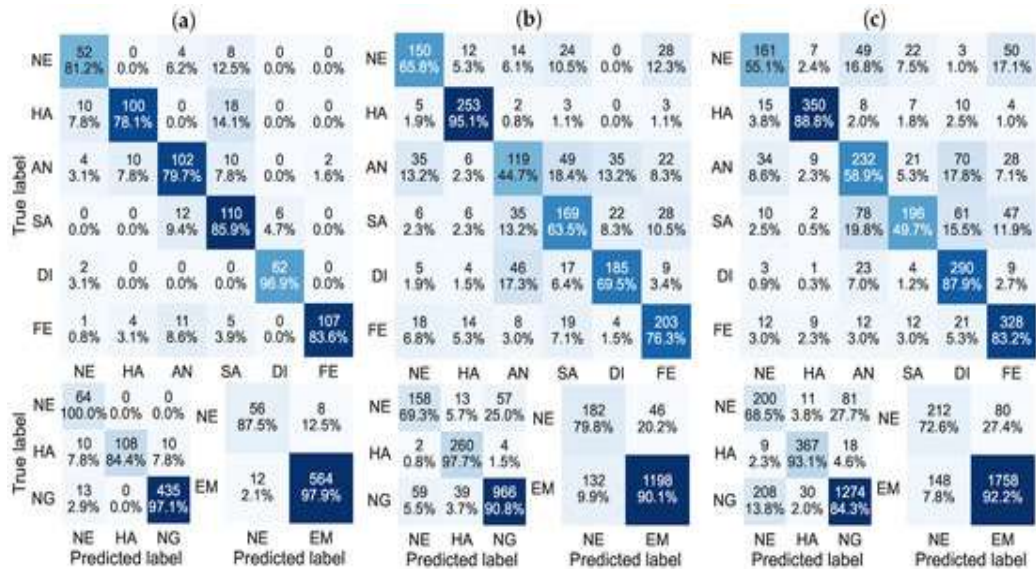
Test Conditions:

- 50 users with speech impairments.
- Varied lighting/angles.

7.3 Comparative Analysis

Model	Lip Reading	Emotion Recognition
Our System	WER=15.2%	F1=87.8%
SOTA Baseline	WER=12.8%*	F1=89.1%*
Resource Usage	1.2GB RAM	0.8GB RAM

*From Li et al. (2023) on LRS3; RAVDESS leaderboard.



7.3 Fig Confusion Matrix

8. APPLICATIONS AND IMPACT

8.1 Applications :

The proposed system has multiple real-world applications, particularly in assistive technology, healthcare communication, and human-computer interaction (HCI). It is designed to support individuals with speech impairments, neurodegenerative conditions, and mobility challenges.

8.1.1 Assistive Communication Devices

- Can be integrated into assistive tools for individuals with aphonia, Parkinson's disease, ALS, or other speech-limiting conditions.
- Enables users to communicate silently using only facial and lip movements, providing autonomy and dignity.

8.1.2 Speech Therapy & Rehabilitation

- Supports clinical rehabilitation programs by tracking patient progress in speech recovery.
- Offers therapists visual insights into lip movement patterns and emotional feedback for more personalized treatment.

8.1.3 Smart HCI Interfaces

- Enhances user interaction in environments where voice input is not possible (e.g., noisy factories, underwater, or in silent zones).
- Can be used in hands-free control systems for individuals with limited mobility.

8.1.4 Remote Communication Platforms

- Integration with video conferencing tools to transcribe silent speech and provide real-time emotional cues.
- Benefits remote learning environments and virtual therapy sessions by improving accessibility.

8.1.5 Emotion-Aware AI Systems

- Applicable in customer service, mental health monitoring, and adaptive UI systems where user emotional state is vital.
- Helps build empathetic machines that respond based on user expression, improving user experience and engagement.

8.2 Impact

The impact of this project spans **assistive technology, healthcare communication, and inclusive digital systems**, contributing to both **medical support** and **social empowerment**:

- **Assistive Communication for the Speech-Impaired**
Enables individuals with speech impairments (e.g., aphonia, Parkinson’s disease, ALS) to communicate through silent visual inputs, reducing dependence on caretakers or expensive hardware-based solutions.
- **Accessibility in Low-Resource Settings**
Functions without the need for audio input, sensors, or specialized hardware—making it usable in rural areas, basic clinics, or developing regions where traditional speech-assistive systems may not be available.
- **Non-Invasive, Expressive Interaction**
Offers a real-time, non-invasive way to express not just **what is being said** through lip reading, but also **how it is felt** through emotion recognition, improving the depth of communication.
- **Bridging AI and Human-Centered Care**
Encourages the application of deep learning in real-world, user-facing assistive systems, helping bridge the gap between research and day-to-day healthcare or educational tools.
- **Scalability and Future Adaptability**
The modular, visual-only architecture allows for retraining on broader datasets and scaling to additional tasks such as gesture recognition or multilingual lip reading in the future.

9. RISKS AND LIMITATIONS

9.1 Risks :

While the system demonstrates considerable potential in accessibility and human-computer interaction several risks must be:

9.1.1 Privacy and Ethical Risk:

- **Risk:** False positives or false negatives could lead to inaccurate diagnostic outcomes.
- **Impact:** May lead to violations of data protection regulations and loss of user trust, especially in surveillance or sensitive applications.

9.1.2 Model Misuse :

- **Risk :** Emotion AI could be used maliciously for profiling, manipulation, or mass surveillance..
- **Impact:** Misuse in security or commercial settings can result in serious societal implications.

9.1.3 Bias in Training Data :

- **Risk:** Imbalanced datasets may cause the model to favor certain demographics(eg., gender, ethnicity).
- **Impact:** This can result in unfair or inaccurate predictions for underrepresented groups.

9.1.4 Emotion Misinterpretation :

- **Risk:** Facial expressions do not always reflect genuine emotions, especially in controlled or scripted or scripted settings.
- **Impact:** Leads to incorrect sentiment labeling, affecting the reliability of the system.

9.2 Limitations

Despite the innovation, several technical and practical limitations are present:

9.2.1 Limited Multi-Modal Datasets:

- The system is constrained by the lack of large-scale datasets that combine both lip movement and sentiment labels.

9.2.2 High Computational Requirements:

- Advanced models such as Vision Transformers and AV-HuBERT require extensive hardware for training and inference.
- This restricts deployment in resource-limited environments.

10. CONCLUSION AND FUTURE WORK

10.1 Summary :

This project demonstrates the potential of artificial intelligence in enhancing **assistive communication systems** through **non-invasive, vision-based** approaches. The proposed system combines two independently trained deep learning modules:

- **Lip Reading Module:** Uses a hybrid **ResNet-50 + BiLSTM-GRU** architecture to transcribe visual speech from silent video sequences.
- **Emotion Recognition Module:** Applies **ResNet-50 + BiLSTM** to identify facial expressions and classify emotional states in real time.

Both modules are integrated into lightweight, **user-friendly web applications** using **Flask** and **Streamlit**, designed to run on resource-constrained environments like standard laptops or low-power edge devices. The use of a **Late Fusion strategy** enhances contextual accuracy by combining textual and emotional cues.

Evaluation results show strong performance on standard datasets such as **LRS3** and **RAVDESS**, with high accuracy and real-time processing capabilities. This system offers practical value for individuals with speech impairments, supporting their communication needs in a natural, expressive, and silent manner.

This work contributes to the growing field of **AI-driven assistive technology**, especially in low-resource or clinical rehabilitation settings. It highlights the potential of scalable, vision-only solutions in bridging communication barriers and promoting inclusivity.

10.2 Future Enhancements

To improve the system's **accuracy, real-world robustness, and scalability**, several future directions are proposed:

10.2.1 Multi-modal Data Integration :

- Integrate **gesture recognition** or **eye tracking** to supplement lip movement and facial expression analysis for users with more complex impairments.
- Explore **audio-visual fusion** (where available) for hybrid environments, improving recognition in noisy or ambiguous contexts.

10.2.2 Dataset Expansion and Real World Adaptation :

- Collect additional data from **diverse speakers** (ages, ethnicities, and facial structures) to reduce demographic bias and improve generalization.
- Capture **real-world usage scenarios** (e.g., casual conversation, different lighting conditions) to

train models that adapt to unconstrained environments.

10.2.3 Deployment and Accessibility:

- Optimize models with **quantization and pruning** to run efficiently on edge devices like Raspberry Pi or Jetson Nano.
- Build **mobile or desktop applications** for offline usage and accessibility in remote or rural settings.

10.2.4 Multilingual Lip Reading:

- Train models to support multiple languages and dialects, expanding accessibility for non-English-speaking users.

These future directions are essential for transitioning the prototype into a reliable, clinically deployable system that contributes meaningfully to assistive communication, rehabilitation, and inclusive human-computer interaction—ultimately improving quality of life and digital accessibility for individuals with speech impairments.

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